

Berka Bank Dataset

Using Data to Spot Risky Loans

Urbanus Kathitu | GitHub: github.com/KathituCodes

Words used in this document

Term	What it means
Query	A question written in a language called SQL and asked of a database. The database answers with a table of results.
Default	When a borrower does not repay a loan as agreed.
Default rate	The share of loans that go bad. If 10 out of 100 loans go bad, the default rate is 10%.
Average balance	The typical amount of money sitting in an account over time.
Inflow	Money coming into an account, such as salary or deposits.
Threshold	A cut-off number. Anything above it gets flagged for a closer look.
Flag	A warning marker placed on an account so a person can review it.

About the data

The Berka dataset contains real records from a Czech bank between 1993 and 1998. It is made up of eight linked tables, a bit like eight connected spreadsheets. Together they follow the whole story of a bank customer: signing up, opening an account, making everyday transactions, taking a loan, and finally repaying it or failing to.

Where the data comes from: <https://www.kaggle.com/datasets/marceloventura/the-berka-dataset>

Table	Rows	What it holds
account	4,500	Every bank account. This is the centre of the dataset.
client	5,369	Every customer.
district	77	Regional facts such as population, average salary and crime rate.
disp	5,369	The link that connects each customer to their account.
loan	682	Every loan issued, including its outcome (called "status"). This is what I wanted to understand.
order	6,471	Standing payment instructions, such as monthly rent.
trans	1,056,320	Every single transaction. This is the largest and most important table.

Table	Rows	What it holds
card	892	Credit cards issued to customers.

Before asking any questions, I read the project description <https://webpages.charlotte.edu/mirsad/itcs6265/group1/index.html> to understand what the data was collected for. I always do this first. Asking questions of data you do not yet understand can produce answers that look convincing but are wrong.

How the loans turned out

Every loan in the data has a status showing how it ended up. I counted how many loans fell into each group before making any assumptions.

Status	Loans	What it means
C	403	Still being repaid, no problems so far. Currently good.
A	203	Fully repaid on time. A good client.
D	45	Still being repaid, but the client is already in financial trouble. Currently bad.
B	31	Loan ended, but the client did not repay everything. A bad client.

To keep things simple, I merged these four statuses into two groups: **GOOD** (A or C) and **BAD** (B or D). Credit scoring models in real banks frame the problem the same way.

The core problem: Out of 682 loans, 606 are good or currently fine and 76 are bad or in trouble. The challenge is to spot the 76 risky borrowers before the bank lends them money.

Making sense of the transactions

The transaction table has over one million rows, and its labels are in Czech. I translated the key terms into English before writing any questions. This mattered: using the wrong label would have returned wrong answers without showing any error.

Czech label	What it means in English	Transactions
VYDAJ + VYBER	Cash withdrawal (money going out)	418,252
PREVOD NA UCET	Transfer to another account (sending money out)	208,283
PRIJEM (no operation name)	Money coming into the account	183,114
VKLAD	Cash deposit	156,743
PREVOD Z UCTU	Transfer from another account (receiving money)	65,226

Czech label	What it means in English	Transactions
VYBER	Cash withdrawal (a second withdrawal label)	16,666
VYBER KARTOU	Withdrawal using a card	8,036

Question 1: Do customers who keep more money in their account default less?

Why is this question important?

Every lender asks this before building a scoring system. If a customer consistently keeps a healthy balance, are they less likely to fail to repay a loan? I used all 1,056,320 transactions to find out.

Answer

1. I worked out the average balance for every account across all of its transactions.
2. I ranked all accounts from lowest to highest average balance and split them into five equal groups. Group 1 is the poorest 20% and Group 5 is the richest 20%.
3. I kept only the lowest 20% and the highest 20%, so the comparison would be clear.
4. I linked these accounts to the loan records and calculated what share of loans went bad in each group.

Results

Group	Total loans	Bad loans	Default rate
Lowest 20% by balance	40	15	37.50%
Highest 20% by balance	246	7	2.85%

Key finding: Accounts with the lowest balances went bad at a rate about 13 times higher than accounts with the highest balances (37.50% divided by 2.85% is about 13).

Insights

- If the bank lends to someone who usually has very little money in their account, about 37 out of every 100 loans go bad. That is a very heavy loss for any lender.
- If the bank lends to someone who usually keeps a high balance, only about 3 out of every 100 loans go bad.
- The gap is not small. Looking at just one everyday habit, the average balance, cuts the risk by a factor of 13. This makes average balance a strong candidate for use in a credit scoring model.
- The top group also had far more loans (246 compared with 40). This suggests the bank was already favouring customers with strong balances, even without a formal scoring system.

Why this matters in the real world

Fintech companies such as Pezesha and M-KOPA use mobile money activity to estimate how much money customers typically keep, even when those customers have no bank account. The idea is the same: steady positive balances are a good sign of reliability. Testing a signal like this is what happens before any scoring model is built.

Technical note: this query used chained CTEs (temporary working tables) and the NTILE(5) function to split accounts into five equal groups.

Question 2: Does a sudden drop in account activity signal trouble?

Why is this question important?

If a customer has a loan and suddenly stops using their account, it can be an early warning. They may have lost their job, their business may have failed, or they may be avoiding the account on purpose. I looked for every account whose number of transactions in a month fell by more than 40% compared with the month before.

Answer

1. I counted the transactions for each account in each month. The dates were stored in a compact code (YYMMDD), so I converted them into a proper year and month first.
2. For each account and month, I looked back and picked up the previous month's count, so that the two months sit side by side.
3. I calculated the percentage change from one month to the next.
4. I marked any drop larger than 40% as HIGH RISK and listed the biggest drops first.

Sample results: the five biggest drops

Account	Year	Month	This month	Last month	Change	Flag
10857	1998	December	1	10	-90.00%	HIGH RISK
10788	1996	March	1	7	-85.71%	HIGH RISK
10019	1998	December	1	7	-85.71%	HIGH RISK
3825	1997	March	1	6	-83.33%	HIGH RISK
573	1998	February	3	18	-83.33%	HIGH RISK

Insights

- **December 1998 kept appearing, but it is a false alarm.** It is the last month in the dataset, and the data stops partway through the month. Accounts look quiet simply because the records ended early, not because customers changed their behaviour. In a real setting, I would leave out the most recent month before flagging anyone.
- The worst cases went from 10 or more transactions in one month to just 1 in the next. A drop of that size deserves a review by a credit officer whatever cut-off is used.

- One query compared every account, month by month, across all 1,056,320 rows in a single pass. This is far faster than checking accounts one at a time, and it is how real banking systems handle this type of analysis.

Why this matters: Spotting that December 1998 was a data problem and not a real risk signal is the kind of judgement that separates a careful analyst from someone who simply reports whatever the numbers show.

Why this matters in the real world

Most lenders assess a borrower once, at the time of application, and then wait to see whether payments are missed. By then the damage is done. Watching for a fall in account activity picks up trouble earlier, giving the lender time to step in before a payment is missed.

Technical note: this query used the LAG() window function to bring each account's previous month next to the current month.

Question 3: Can we spot customers pulling out their money before they default?

Why is this question important?

When someone knows they cannot repay a loan, a common pattern is to take out as much money as possible before they stop paying. This is known as cash-out before default, and it is a well-studied form of lending fraud. I looked for accounts that met both of these conditions at the same time:

- **Condition 1:** the account has a bad or troubled loan (status B or D).
- **Condition 2:** the most recent transaction was a large payment out, bigger than a set share (first 80%, then 50%) of the account's average monthly income.

Either condition alone is not a strong warning. Together, they may point to money being moved out on purpose.

My first attempt failed, and this is what I learned

My first version found each account's latest transaction by checking the whole table again for every single row. With over a million rows, that meant more than a million separate checks. The query ran for over an hour and I cancelled it.

I rewrote it so that the database ranks each account's transactions from newest to oldest in one pass, then keeps only the newest. This finished in seconds.

Lesson: A query that gives the right answer but takes an hour is not ready to be used in the real world.

Answer

1. I calculated each account's average monthly income, counting only money coming in.
2. I ranked every transaction by date, newest first, so that rank 1 is the latest transaction.
3. I kept only rank 1, which gives one row per account: its most recent transaction.

4. I listed all accounts with a bad or troubled loan.
5. I combined these lists and flagged accounts where the latest payment out was bigger than the chosen share of average income.

Comparing two cut-offs: 80% and 50%

I ran the query twice with different cut-offs to see how the results change. A credit team faces the same choice when deciding how strict its warning system should be.

Account	Loan amount	Status	Last payment out	Avg monthly inflow	% of inflow	At 80% cut-off	At 50% cut-off
10857	385,560	D	6,426.30	8,721.80	73.68%	MONITOR	HIGH RISK
2176	299,088	B	1,085.00	13,765.84	7.88%	MONITOR	MONITOR
3678	316,140	D	42.80	5,047.83	0.85%	MONITOR	MONITOR
1888	57,360	B	36.20	10,192.21	0.36%	MONITOR	MONITOR
3037	277,884	D	14.60	11,308.64	0.13%	MONITOR	MONITOR
4081	137,904	B	5.50	6,236.46	0.09%	MONITOR	MONITOR
3273	94,512	B	6.80	17,769.74	0.04%	MONITOR	MONITOR
2236	139,464	D	0.40	9,600.54	0.00%	MONITOR	MONITOR

Insights

- **At the 80% cut-off, no account was flagged as high risk.** Everything showed as MONITOR. A credit officer looking at this could wrongly conclude that there is no cash-out risk.
- **At the 50% cut-off, account 10857 was flagged as high risk straight away.** It has the largest troubled loan in the bad group (385,560), a status of D (still running, but the client is already in trouble), and its last transaction was a transfer of 6,426 to another account, which is 73.68% of its average monthly income.
- The fact that this was a transfer to another account, and not an ordinary cash withdrawal, makes it more concerning. It hints at money being moved deliberately, not spent as usual.
- The other seven accounts all made small payments, below 8% of their average income. These look like routine spending and stay on MONITOR whichever cut-off is used.

Key finding: Account 10857 is the only real warning sign under this definition of cash-out before default. It has the biggest loan, the worst status and the largest recent outflow compared with income. A cut-off of 80% misses it completely, while 50% catches it immediately.

The trade-off: catching more risks versus more false alarms

- **A high cut-off (strict):** catches fewer accounts, but those it catches are almost certainly real risks. The downside is that some real risks slip through.
- **A low cut-off (sensitive):** catches more real risks, but also flags good customers by mistake. This wastes review time and can inconvenience honest customers.
- No cut-off is mathematically correct. The right one depends on how much the business values catching bad loans compared with the cost of false alarms. A data scientist provides the evidence, but the decision is a business one.

In practice, I would show both versions to the credit team and let them choose based on how much risk they are willing to accept.

Technical note: this query used the ROW_NUMBER() window function to find each account's latest transaction. In data science, this trade-off is known as precision versus recall.

Summary of all three questions

Question	What I asked	What I found	How a bank would use it
1. Balance and default	Do customers with high balances default less?	Default rate is about 13 times higher in the lowest 20% than in the highest 20%.	Use average balance as an input to a credit score.
2. Drop in activity	Does a sudden fall in activity warn of trouble?	Some accounts dropped 90% in one month. December 1998 results were a data issue.	Monitor borrowers after the loan is paid out, to act early.
3. Cash-out before default	Can we spot money being pulled out before default?	Account 10857 was the only real warning sign. The cut-off decides who gets flagged.	Add a fraud check on top of normal credit scoring.

Together, these three questions cover the full workflow of a fintech data team, testing which signals predict risk, watching customers over time, and detecting possible fraud. Each used a different SQL technique:

- NTILE for Question 1
- LAG for Question 2
- ROW_NUMBER for Question 3

Together they form a complete portfolio of advanced SQL applied to a real financial dataset.